1 Coordinate and law

(bars denote normalized quantities: lengths by R, $\bar{a} = a/a_o$, $\bar{f} = a_o f$; $0 < \bar{a} < 1 \Leftrightarrow u > 1$; $e_\theta = \theta - \hat{\theta}$)

$$u := \frac{1}{1-\bar{a}}, \qquad \bar{a} = 1 - \frac{1}{u}, \qquad a_w = a_o\,\bar{a}_w, \qquad a_o = \frac{\Delta t}{\gamma_N R}$$

$$\frac{\mathrm{d}\bar{a}}{\mathrm{d}\bar{w}} = \big(1-\bar{a}\big)^2\big(1+\delta a\big) \qquad \Longleftrightarrow \qquad \frac{\mathrm{d}u}{\mathrm{d}\bar{w}} = 1 + \delta a$$

$$u(\bar{w}) = \big(1+\delta a\big)\big(\bar{w} - \bar{w}_0\big), \qquad \frac{\partial u}{\partial \bar{w}_0} = -\big(1+\delta a\big), \qquad \int_{\bar{w}[k]}^{\bar{w}[k+1]} \mathrm{d}u = \big(1+\delta a\big)\,\Sigma[k] \quad \text{(exact; } \Sigma \text{ in S3)}$$

2 Jacobians

$$\frac{\partial \bar{a}}{\partial u} = \frac{1}{u^2} = \big(1-\bar{a}\big)^2, \qquad \frac{\partial}{\partial u}\Big[u + (1+\delta a)\big(\,\cdot\,\big)\Big] = 1, \qquad \frac{\partial}{\partial\,\delta a}\Big[u + (1+\delta a)\,\Sigma\Big] = \Sigma$$

3 True dynamics

$\big(\delta\bar{w}_j[k] := \delta\bar{w}[k-d+j-1],\ j = 1,\dots,d+1,\ d = 2\big)$

$$\begin{aligned} \bar{w}[k] &= \bar{w}_d[k] - \delta\bar{w}_3[k], & \bar{w}_d[k+1] &= \bar{w}_d[k] + \Delta\bar{w}_d[k] \\ \delta\bar{w}_1[k+1] &= \delta\bar{w}_2[k], & \delta\bar{w}_2[k+1] &= \delta\bar{w}_3[k] \\ \delta\bar{w}_3[k+1] &= \lambda_c\,\delta\bar{w}_3[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \varepsilon_w[k] \end{aligned}$$

$$\begin{aligned} F_{dw}[k] &= \bar{f}_{dw}[k] + (1-\lambda_c)\sum_{i=1}^{d} \bar{f}_{dw}[k-i] \\ \varepsilon_w[k] &= \bar{a}_w[k]\,\bar{f}_T[k] + (1-\lambda_c)\sum_{i=1}^{d} \bar{a}_w[k-i]\,\bar{f}_T[k-i] + (1-\lambda_c)\,n_w[k] \end{aligned}$$

$$\Sigma[k] := \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\bar{w}_3[k] + F_{dw}[k]\,e_{\bar{a}_w}[k] + \varepsilon_w[k] \;=\; \bar{w}[k+1] - \bar{w}[k]$$

$$\text{(a)} \quad u[k+1] = u[k] + \Sigma[k], \qquad \text{(b)} \quad u[k+1] = u[k] + \big(1+\delta a[k]\big)\Sigma[k], \qquad \delta a[k+1] = \delta a[k]$$

4 Process model

$$\text{(a)} \quad \mathbf{x} = \big[\, \delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ u \,\big]^\top, \qquad \text{(b)} \quad \mathbf{x} = \big[\, \delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ u,\ \delta a \,\big]^\top \qquad (d=2)$$

5 Estimator

$$\hat{\Sigma}[k] := \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k]$$

$$\delta\hat{\bar{w}}_1[k+1] = \delta\hat{\bar{w}}_2[k], \qquad \delta\hat{\bar{w}}_2[k+1] = \delta\hat{\bar{w}}_3[k], \qquad \delta\hat{\bar{w}}_3[k+1] = \lambda_c\,\delta\hat{\bar{w}}_3[k]$$

$$\text{(a)} \quad \hat{u}[k+1] = \hat{u}[k] + \hat{\Sigma}[k], \qquad \text{(b)} \quad \hat{u}[k+1] = \hat{u}[k] + \big(1+\delta\hat{a}[k]\big)\hat{\Sigma}[k], \qquad \delta\hat{a}[k+1] = \delta\hat{a}[k]$$

$$\hat{\bar{a}}_w[k] = 1 - \frac{1}{\hat{u}[k]} \qquad \text{(control law and S7 consume this)}$$

6 Error dynamics

(True $-$ Estimator; coefficients at the estimate; $e_{\bar{a}_w} = \big(1-\hat{\bar{a}}_w\big)^2 e_u + \mathcal{O}(e_u^2)$)

$$\begin{aligned}
e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k], \qquad e_{\delta\bar{w}_2}[k+1] = e_{\delta\bar{w}_3}[k] \\
e_{\delta\bar{w}_3}[k+1] &= \lambda_c\, e_{\delta\bar{w}_3}[k] - F_{dw}\big(1-\hat{\bar{a}}_w\big)^2 e_u[k] - \varepsilon_w[k] \\
e_u[k+1] &= \Big[\,1 + \big(1+\delta\hat{a}\big)F_{dw}\big(1-\hat{\bar{a}}_w\big)^2\Big] e_u[k] + \hat{\Sigma}[k]\, e_{\delta a}[k] \\
&\quad + \big(1+\delta\hat{a}\big)(1-\lambda_c)\, e_{\delta\bar{w}_3}[k] + \big(1+\delta\hat{a}\big)\varepsilon_w[k] \\
e_{\delta a}[k+1] &= e_{\delta a}[k]
\end{aligned}$$

$$\text{(b)} \quad F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw}(1-\hat{\bar{a}}_w)^2 & 0 \\ 0 & 0 & (1+\delta\hat{a})(1-\lambda_c) & 1+(1+\delta\hat{a})F_{dw}(1-\hat{\bar{a}}_w)^2 & \hat{\Sigma} \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

(a) is the leading 4×4 block with $\delta\hat{a} = 0$.

$$\mathbf{q}[k] = \big[\, 0,\ 0,\ -\varepsilon_w,\ \big(1+\delta\hat{a}\big)\varepsilon_w,\ 0 \,\big]^\top$$

7 Measurement and innovation

$$\begin{aligned}
\nabla_d \bar{w}_d[k] &= \bar{w}_d[k] - \bar{w}_d[k-d] \\
y_1[k] &= \delta\bar{w}_1[k] + n_w[k] \\
y_2[k] &= \bar{a}_w[k-d] + n_a[k]
\end{aligned}$$

$$u[k-d] = u[k] - \big(1+\delta a\big)\nabla_d \bar{w}_d[k] + \big(1+\delta a\big)\big(\delta\bar{w}_3[k] - \delta\bar{w}_1[k]\big)$$

$$\hat{v}[k] := \hat{u}[k] - \big(1+\delta\hat{a}\big)\nabla_d \bar{w}_d[k] \quad \big[\text{declared truncation: the } (\delta\bar{w}_3 - \delta\bar{w}_1) \text{ channel is dropped from } H\big]$$

$$\hat{y}_2[k] = 1 - \frac{1}{\hat{v}[k]}, \qquad \frac{\partial y_2}{\partial u} = \frac{1}{\hat{v}^2}, \qquad \frac{\partial y_2}{\partial\,\delta a} = -\frac{\nabla_d \bar{w}_d}{\hat{v}^2}$$

$$\text{(b)} \quad H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \hat{v}^{-2} & -\nabla_d \bar{w}_d\,\hat{v}^{-2} \end{bmatrix}, \qquad K[k] = P^f H^\top \big(H P^f H^\top + R\big)^{-1}$$

$$\begin{aligned}
e_{y_1}[k] &= e_{\delta\bar{w}_1}[k] + n_w[k] \\
e_{y_2}[k] &= \hat{v}^{-2}\, e_u[k] - \nabla_d \bar{w}_d\, \hat{v}^{-2}\, e_{\delta a}[k] + n_a[k]
\end{aligned}$$

8 Process and measurement noise

$$q_3 = -\bar{a}_w\,\bar{f}_T, \qquad q_u = -\big(1+\delta\hat{a}\big)q_3, \qquad \gamma(\bar{w}) = \gamma_N c(\bar{w}), \qquad \sigma^2_{\bar{f}_T} = a_o^2\,\frac{4k_B T\gamma(\bar{w})}{\Delta t}$$

$$Q_{33} = \frac{4k_B T\,a_o}{R}\,\bar{a}_w, \qquad Q_{3u} = Q_{u3} = -\big(1+\delta\hat{a}\big)Q_{33}, \qquad Q_{uu} = \big(1+\delta\hat{a}\big)^2 Q_{33}, \qquad Q_{\delta a} = 0$$

$$R = \mathrm{diag}(R_1, R_2), \qquad R_1 = \sigma^2_{n_w}, \qquad R_2 = K_{\mathrm{var}} IF_{\mathrm{var}}\big(\bar{a}_w + \xi\big)^2 + d\,\hat{v}^{-4} Q_{uu}, \qquad K_{\mathrm{var}} = \frac{2a_{\mathrm{cov}}}{2 - a_{\mathrm{cov}}}$$

$$\sigma^2_{\delta\bar{w}_r} = C_{dpmr}\frac{4k_B T\,a_o}{R}\big(\bar{a}_w + \xi\big), \qquad \xi = \frac{C_n \sigma^2_{n_w}}{C_{dpmr}\big(4k_B T a_o/R\big)}$$

Seed and prior

$$\delta\hat{a}[0] = 0, \qquad \hat{u}[0] = \bar{w}[0] - \hat{\bar{w}}_0 = \frac{1}{1 - \hat{\bar{a}}_w[0]}$$

$$\frac{\partial u}{\partial \bar{w}_0} = -\big(1 + \delta a\big), \qquad \left.\frac{\partial u}{\partial\, \delta a}\right|_{\bar{w}} = \bar{w} - \bar{w}_0 = \frac{u}{1 + \delta a}, \qquad \delta u = \frac{\delta \bar{a}}{\big(1 - \bar{a}\big)^2} = u^2\, \delta \bar{a}$$

$$P_{uu}[0] = \left(\frac{\hat{u}[0]}{1 + \delta\hat{a}}\right)^2 P_{\delta a} + \big(1 + \delta\hat{a}\big)^2 P_{\bar{w}_0} + \hat{u}[0]^4\, \mathrm{floor}_a^2$$

$$P_{u\,\delta a}[0] = \frac{\hat{u}[0]}{1 + \delta\hat{a}}\, P_{\delta a}, \qquad P_{\delta a}[0] = P_{\delta a}, \qquad u \geq u_{\min}$$